

Asandei Ștefan-Alexandru

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EDUCATION

National College of Iasi

High School, Computer Science and Mathematics

Sep 2022 — May 2026

Iasi, Romania

COMPETITIONS

International Olympiad in Artificial intelligence, Bronze Medal

China, Aug 2025

- Most important AI competition for high school student, 2nd edition
- 2 days of contest with 6 tasks, with a combination of tabular, NLP and CV tasks

National Olympiad in Artificial intelligence, 2nd Place, Gold Medal

May 2025

- Solved tasks from the domains of Natural Language Processing and Computer Vision
- Qualified to the selection camp for IOAI

FTC World Championship, "Motivate" Award Finalist

Texas, USA, May 2023

- Robotics competition where students work in teams to build a robot for a given challenge

National Olympiad of Digital Innovation, 1st Place & Gold Medal

Romania, May 2024

- Project-based olympiad, built web applications to utilize innovative systems in smart ways
- Hackathon project was to make an app like Waze, but using only free OSS tools
- I used OpenStreetMap along with an ML microservice (YOLO, OpenCV) to monitor surveillance cameras in the city for live traffic data

National Olympiad of Informatics, Bronze Medal

Romania, May 2024

National Olympiad of Informatics in Teams, Bronze Medal

Romania, February 2024

Romanian Science & Engineering Fair, 3rd Place

Romania, May 2024

Romanian CanSat and Rocketry Championship, 3rd Place

Romania, May 2024

- Team competition to build & launch a CanSat that solves a given telemetry challenge
- Wrote the code for the embedded system in our CanSat using the AVR toolchain

PROJECTS

Explained IOAI Tasks (github.com/stefanasandei/roai-solved)

Apr 2025 — Present

- Collection with IOAI-level tasks solved and explained, also translated national-level contests to english
- Contests included: National Selection (Romania, Poland, Northern Eurasia, Asia Pacific), International (IOAI, IAIO)

Orion - Personal Knowledge Base web app (orion.asandei.com)

Dec 2024 — Aug 2025

- Tech stack: Frontend (Svelte, Tailwind, Typescript, ShadCN UI), Backend (SvelteKit, tRPC, Drizzle, PostgreSQL), Agentic AI (Langchain, Vercel AI SDK, pgvector)
- Experimented with various RAG techniques to efficiently interact with the PKB

Steer - Research to explore methods for self-driving models (steer.asandei.com)

Jul 2024 — Oct 2024

- Used Mamba2 (state space model) as a backbone for a self-driving model
- Benchmarked the performance against PilotNet (CNN based) and Seq2Seq (RegNet+GRU based)
- Explained findings in a technical report

Helix - Online Judge coding platform (helix.asandei.com)

Apr 2023 — Aug 2023

- Tech stack: Frontend (React, Tailwind, Typescript), Backend (NextJS, tRPC, Prisma, PostgreSQL, Rust)
- Wrote a custom (very efficient) code execution engine, that utilized Linux namespaces for lightweight sandboxing

VOLUNTEER EXPERIENCE

Problem setter, Organizer

Sep 2025 — Present

IOAI Community Contest

- Started a monthly contest, to help students better prepare for the IOAI
- Designed and implemented challenging tasks, along with explanations and learning resources

Tutor

Sep 2025 — Present

IOAI Community

- Held free tutoring session for high school students wanting to prepare for the IOAI
- Explained concepts from Machine Learning, Deep Learning and the related Mathematics

CERTIFICATES

Cambridge Advanced English, C2 with 200 overall score

Dec 2024

SKILLS

- **Programming Languages:** Python, C++, Bash, Go, Typescript
- **Technologies:** Docker, vLLM, SGLang, PostgreSQL, Redis, RabbitMQ, gRPC
- **Libraries/Frameworks:** Python Libraries (NumPy, PyTorch, JAX, Scikit-Learn, Triton, etc.), C++ (CMake, CUDA, CUTLASS, Vulkan, Metal, ImGui, etc.) and some Web libraries (React, Tailwind, NextJS, SvelteKit, etc.)
- **Developer Tools:** Git, Vim, Amazon Web Services